



Form Filler Agent

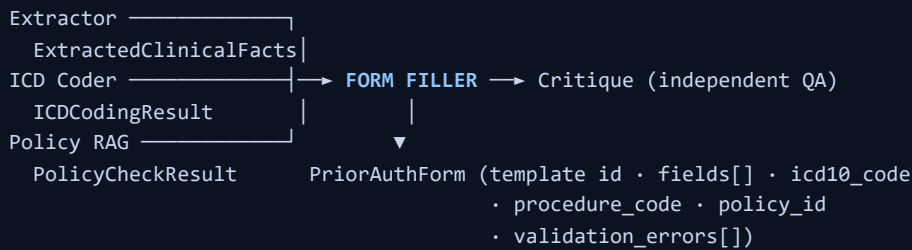
Architecture Document — Template-Driven Form Assembly (Current)

The complete technical architecture of Agent 4 in the Prior Authorization pipeline:
fuzzy template selection, declarative field builders, dual-layer validation
(completeness + cross-agent consistency), and a fully deterministic, LLM-free
design.

System	Prior Authorization AI — multi-agent workflow
Component	Agent 4 · Form Filler (<code>agents/form_filler.py</code>)
Approach	Deterministic template assembly — no LLM, no external API
Template library	9 insurer form templates · 9 field types (union)
Inputs	Outputs of Extractor + ICD Coder + Policy RAG (all Pydantic-typed)
Output	<code>PriorAuthForm</code> — validated, audit-ready, verified downstream by the Critique Agent
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Where the agent sits, and what it owns

The Form Filler is the fourth of five agents — the **assembler**. Everything before it produces evidence (facts, diagnosis code, policy verdict); the Form Filler converts that evidence into the artifact an insurer actually receives: a completed, validated prior authorization form. It is the only agent that consumes *all three* upstream outputs at once.



Its design invariants:

Invariant 1 — Assemble, never invent

Every field value is copied or mechanically derived from an upstream agent's typed output. The Form Filler contains no AI and no free-text generation: a value it cannot source is reported as a validation error — never fabricated.

Invariant 2 — Validation is the product

The form carries its own defect list (`validation_errors[]`). A structurally complete form with unresolved conflicts still flows onward — but the orchestrator reads `form.is_valid` and routes the case to `FAILED_VALIDATION` instead of approval.

The four stages inside the agent

FORM FILLER AGENT

agents/form_filler.py

STAGE 1 • TEMPLATE SELECTION

requested_procedure vs. each template's procedure_keywords
score = 1.0 on substring containment, else difflib ratio
floor: best score < 0.30 → VALIDATION_FAILED handoff (no template)
9 templates: TKR • THR • ortho-general • cardiac • oncology • ...

| winning template (required_fields[])

STAGE 2 • FIELD ASSEMBLY (declarative builders)

patient_age / patient_sex / laterality ← ExtractedClinicalFacts
diagnosis_code ← ICDCodingResult
policy_id ← PolicyCheckResult
procedure_code = "PROC-" + policy_id suffix (derived, checkable)
failed_treatments → "None documented" when legitimately empty
imaging_evidence → "Not provided" when legitimately absent
unsourceable value → literal "UNSPECIFIED" (never silently blank)

| fields[] (name • value • required flag)

STAGE 3 • VALIDATION (two independent nets)

Completeness: hard-required fields must carry real values
{patient_age, patient_sex, diagnosis_code, procedure_code,
policy_id} → error per unfilled hard-required field
laterality == "unknown" → error (ambiguity is never submitted)
Cross-agent consistency:
X note laterality ≠ requested-procedure laterality
X ICD-10 code laterality ≠ documented diagnosis laterality

| validation_errors[]

STAGE 4 • TYPED OUTPUT

PriorAuthForm (Pydantic-validated)
confidence = 1.0 if clean, else 1.0 - 0.15 × error_count (floor 0)

|
| form.is_valid & policy met → COMPLETED (auto-approval)
| form.is_valid, policy unmet → SUSPENDED_POLICY_MISMATCH → human
| validation errors → FAILED_VALIDATION
(then Agent 5 • Critique re-verifies everything)

3 • COMPONENT INVENTORY

Every moving part, file by file

COMPONENT	LOCATION	RESPONSIBILITY
Agent shell	agents/form_filler.py	Template selection · field assembly · dual validation · confidence · typed Handoff[PriorAuthForm]
Template library	dataset/legacy_full/form_templates/templates.json	9 insurer form templates: template_id , procedure_keywords , required_fields (8 each, from a 9-field vocabulary)
Field builders	_field_builders()	Declarative map field_name → lambda pulling from the right upstream payload — adding a new form field is one dictionary entry
Fuzzy matcher	difflib.SequenceMatcher	Template scoring: containment = 1.0, otherwise string-similarity ratio; _MIN_TEMPLATE_SCORE = 0.3 floor
Hard-required set	_HARD_REQUIRED_FIELDS	The five fields a form cannot be submitted without; informational fields (treatments, imaging) may be legitimately "empty"
Output contract	schemas/form_output.py	PriorAuthForm + FormField ; is_valid property = len(validation_errors) == 0
Downstream verifier	agents/critique.py	Independently re-checks every form value against upstream agents (age, sex, ICD, policy, procedure-code derivation) after assembly

The nine field types and their sources

FIELD	SOURCE OF TRUTH	EMPTY-VALUE POLICY
patient_age	Extractor · facts.age	hard-required → error if missing
patient_sex	Extractor · facts.sex	hard-required → error if missing
laterality	Extractor · facts.laterality	"unknown" → validation error (ambiguity blocks)
diagnosis_code	ICD Coder · icd10_code	hard-required → error if missing
procedure_code	Derived: "PROC-" + policy_id suffix	hard-required; derivation re-verified by Critique
policy_id	Policy RAG · policy_id	hard-required → error if missing
failed_treatments	Extractor · joined list	informational → "None documented" (a legitimate signal)
imaging_evidence	Extractor · free text	informational → "Not provided" (a legitimate signal)
human_review_flag	Constant "false"	flipped by humans downstream, never by the AI

4 · RUNTIME SEQUENCE

What happens on every case, in order

#	STEP	DETAIL	FAILURE BEHAVIOUR
1	Select template	Score all 9 templates against requested_procedure ; containment beats fuzzy ratio	best < 0.30 → VALIDATION_FAILED handoff; case ends as FAILED_VALIDATION
2	Derive procedure code	"PROC-" + policy_id.split("-")[-1] — deterministic, so the Critique Agent can re-derive and verify it	—
3	Assemble fields	Run the builder lambda for each of the template's required_fields ; missing builder → empty → "UNSPECIFIED"	hard-required gaps recorded as errors, not exceptions
4	Completeness check	Five hard-required fields must be non-empty; ambiguous laterality rejected	appends to validation_errors[]
5	Consistency check	Note-vs-procedure laterality; ICD-code-vs-note laterality	appends to validation_errors[]
6	Emit typed form	PriorAuthForm Pydantic-validated; confidence scored; OK handoff (defects travel inside the form)	orchestrator routes on form.is_valid

confidence = 1.0 (clean form) · otherwise max(0, 1.0 - 0.15 × validation_error_count)

Worked example (auto-approved TAVR case)

```
requested_procedure = "TAVR"
template scores → FORM-CARD-04 (cardiac) wins · others < floor
procedure_code → policy POL-CARD-TAVR-011 → "PROC-011"
fields → age 72 · sex F · diagnosis_code I35.0 · policy POL-CARD-TAVR-011
        imaging "Echocardiogram confirms..." · treatments "None documented"
validation → 0 errors → is_valid = true → confidence 1.00
orchestrator → policy criteria met → COMPLETED (auto-approved)
Critique (5th) → re-verifies every value independently → PASS · 100/100
```

5 · FAILURE & SAFEGUARD MATRIX

Defect handling

CONDITION	BEHAVIOUR	CASE OUTCOME
No template scores ≥ 0.30	Failed handoff with the unmatched procedure named	FAILED_VALIDATION
Hard-required field unfillable	Field emitted as "UNSPECIFIED" + validation error	FAILED_VALIDATION
Laterality "unknown" (note said both sides)	Explicit validation error — never guesses a side	FAILED_VALIDATION
Note laterality ≠ procedure laterality	Consistency error (belt-and-braces: the orchestrator also gates this earlier)	FAILED_VALIDATION
ICD code laterality ≠ documented laterality	Consistency error sourced from the ICD Coder's laterality_match flag	FAILED_VALIDATION
Form clean but policy criteria unmet	Form still produced (evidence preserved for the reviewer)	SUSPENDED_POLICY_MISMATCH → human queue

Why defects travel inside the form (not as exceptions)

A hospital reviewer fixing a rejected case needs to see the *whole* form with the exact gaps marked — not a stack trace. Emitting a complete-but-annotated form preserves every piece of assembled evidence for the human who picks the case up.

6 · SECURITY & AUDITABILITY

Trust properties

- **Deterministic end-to-end:** no LLM, no network, no randomness — identical inputs always produce byte-identical forms.
- **PHI:** operates exclusively on the already-masked payloads; patient identifiers cannot reach a form field by construction.
- **Traceable values:** every field's provenance is a typed upstream output; the `procedure_code` derivation is re-computable by auditors and is re-computed by the Critique Agent on every run.
- **Independent QA:** Agent 5 re-validates form-vs-Extractor, form-vs-ICD, form-vs-Policy and the decision itself — a second pair of eyes that shares no code with the assembler.
- **Extensible by data:** new insurer forms are added by appending a template (keywords + field list) to `templates.json` ; new field types are one builder entry — no logic changes.